# Problem Set 8

Foundations of Machine Learning

- **Submission**: Turn in as a `PDF` and a **separate source code file** on `http://mycourses.unh.edu`. Scanned handwritten notes are encouraged.
- **Questions**: See the class website for a link to office hours

**Relevant reading**: CO 4.3, ILA 10.4 (linear programming); ISL 2–3 (linear regression). Problems marked **[code]** should be submitted as a program in a language of your choice (Julia or Python recommended). The cited problem numbers are from ILA, 5th edition (Strang, 2016), and ISL, 2nd edition (James et al., 2021).

1. ILA, Problem Set 10.4 (linear programming): problems 1, 3.
2. **[code]** Formulate and solve a small revenue-maximization linear program (using, e.g., JuMP in Julia or PuLP/SciPy in Python). Report the optimal solution, the optimal value, and which constraints are active.
3. ISL, Chapter 3, Conceptual exercises: problems 1, 3, 4.
4. ISL, Chapter 3, Applied exercises: the simple linear regression exercise on the `Auto` data set (problem 8) **[code]**.
5. **Time spent.** Report approximately how many hours you spent completing this problem set (including the reading).